KARTHIKEYAN A

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Portfolio: https://karthikeyana-tech.github.io/portfolio/

Professional Summary

Agricultural Engineering undergraduate skilled in **SOLIDWORKS, PTC Creo, and AutoCAD**, with experience in **3D CAD modelling, mechanical design, assembly design, and engineering drawings**. Strong foundation in **GD&T, fits and clearances, and design for manufacturability**, seeking to contribute to product development as a **Design Engineer**.

Technical Skills

CAD & Design: AutoCAD, SolidWorks, Creo, 3D Part Modelling, Assembly Design, GD&T, Tolerance Stack-up
CAE & Simulation: ANSYS Workbench, Static Structural Analysis, Thermal Analysis, Modal Analysis
Core: Farm Equipments, Engineering Mechanics, Strength of Materials, Machine Design, Kinematics, Thermodynamics, Heat Transfer
Engineering Practices: Fits & Clearances, BOM Preparation, Design Documentation, Engineering Drawing Standards, Design Review
Office Tools: Microsoft Excel (Data Entry, Basic Formulas, Tables), Microsoft Word, Microsoft PowerPoint, Google Workspace

Projects

Agricultural Cultivator CAD Design & Assembly* | *SolidWorks ***Oct 2026***

- Modelled the **frame, tines, blades, clamps, three-point hitch, and hardware** as individual 3D CAD components.
- Integrated components into a **complete assembly** and created rendered visualizations and CAD documentation.

Surface Modelling of Agricultural Digging Tool* | *SolidWorks ***Feb 2026***

- Developed an agricultural digging tool using **Trim Surface, Fill Surface, Curvature Continuity, and Thicken**.
- Applied **surface modelling and design for manufacturability (DFM)** principles to develop the tool geometry.

Shock Absorber Assembly Design* | *SolidWorks ***Feb 2026***

- Modelled and assembled a **parametric shock absorber** using sweeps, helix, threading, and mounting components.
- Applied **assembly mates** and PhotoView 360 rendering for assembly validation and visualization.

Four-Cylinder Inline Engine Assembly Design* | *Creo ***Jun 2025***

- Modelled key engine components, including the **crankshaft, pistons, connecting rods, and cylinder block** using Creo Parametric.
- Created a **fully constrained 3D assembly** and validated component motion through animation.

Internship Experience

Python with AI/ML Intern | QuantumNique Solutions Pvt. Ltd. *Jul–Aug 2026*

- Completed a **30-day internship** in **Python with Artificial Intelligence and Machine Learning**.
- Recognized as one of the **Best Performing Students** during the internship.

Quality Control Intern | The Salem District Co-operative Milk Producers' Union Ltd. (Aavin) *Jun–Jul 2025*

- Completed internship training in **dairy quality assurance and quality control** at Salem Aavin.
- Gained practical exposure to **dairy industry operations and quality control practices**.

Education

- **B. Tech. (Honors)** - Agricultural Engineering | Hindustan College of Technology, Salem | **CGPA: 8.01/10** | 2027
- **Class XII (CBSE – AISSCE)** | Sri Vidya Mandir Higher Secondary School, Salem | **Percentage: 77.67%** | 2023
- **Class X (CBSE – AISSE)** | Sri Vidya Mandir Higher Secondary School, Salem | **Percentage: 81.17%** | 2021

Leadership & Responsibilities

- **Innovation Ambassador** | **MoE's** Innovation Cell at Hindusthan College of Technology — Feb 2026 – Present
- **Chairman | SAMRAT** at Hindusthan College of Technology — **Academic Year:** 2025–2026
- **Vice-Chairman | SAMRAT** at Hindusthan College of Technology — **Academic Year:** 2024–2025

Achievements

- **Smart India Hackathon (SIH) 2025 – Grand Finalist (Hardware Edition):** Led **Team AGRO VISIONARIES** to the **National Grand Finale**, representing **Hindusthan College of Technology** at **Manipal University Jaipur, Rajasthan**.
- **Best Performing Individual – National IDE Bootcamp 2026:** Awarded **Best Performing Individual** at the **National-Level Innovation, Design & Entrepreneurship (IDE) Bootcamp**, **JNNCE, Shivamogga, Karnataka**.
- **Best Performing Team – National IDE Bootcamp 2026:** Led **Team AGRO VISIONARIES** to **Best Performing Team** recognition at the **National-Level IDE Bootcamp**; project **featured in *The Indian Express*.**
- **Winner** of **EDII–TNYIEDP Nimirndhu Nil Hackathon 2026.**
- **1st Prize – CAD Challenge, TALENT FORCE 26**: Secured first place among **100+ participants** at the HCT Engineer's Show, organized by the R&D and I&E Cells of Hindusthan College of Technology, Salem.